1. System Overview: How HCMS Operates

The **Health Center Management System (HCMS)**, identified as **e-health.albania**, serves as a centralized digital nervous system for healthcare facilities. In its initial phase, the system acts as a secure data gateway, requiring a **Secure Login** from all personnel to ensure that sensitive medical data is accessed only by authorized actors. Once authenticated, the system facilitates a seamless transition from administrative tasks to clinical care.

It begins with the **Patient Management Module**, where new arrivals are registered and assigned a unique ID. From there, the system coordinates the facility's resources by checking real-time doctor schedules and managing time slots. As the patient moves from the reception desk to the consultation room, the system shifts its focus to the **Medical Records Module**, providing doctors with instant access to patient history and the ability to input new diagnostic data. By automating these routine operations, HCMS reduces administrative bottlenecks and minimizes the risk of human error in clinical documentation.

2. Functional Classification Matrix

To distinguish between interaction, ownership, and benefit, stakeholders are categorized by their primary relationship with the software system.

Classification	Definition	Primary Focus	HCMS Stakeholder Examples
User	Individuals who directly operate the system interface to perform daily organizational tasks.	Operational efficiency and interface usability.	Doctors, Nurses, Receptionists, Administrators.
Client	Entities that commission, fund, and define the strategic requirements of the system.	Strategic alignment, ROI, and regulatory compliance.	HC Director, Board of Directors, Ministry of Health.

Customer	The ultimate beneficiaries of the system's services or data outputs.	Service quality, data accuracy, and health outcomes.	Patients, Insurance Companies, Pharmacies.

3. Stakeholder Onion Diagram Analysis

The stakeholder landscape is organized into concentric rings, representing the level of influence and proximity to the system core.

3.1 The System Core: e-health.albania

The center represents the technical solution itself, acting as the secure repository for clinical and administrative data.

3.2 Ring 1: Operational Stakeholders (Primary Users)

This layer contains the actors who maintain high-frequency interaction with the system.

- **Medical Staff:** Doctors and Nurses who record diagnoses, prescriptions, and update patient vitals.
- **Administrative Staff:** Receptionists who facilitate patient registration and appointment scheduling.

3.3 Ring 2: Internal Stakeholders

These individuals operate within the organization, focusing on the system's infrastructure and backend utility.

- **Technical Support & HR:** Responsible for troubleshooting, infrastructure deployment, and managing staff records.
- **Accounting:** Monitors billing and financial data generated by clinical operations.

3.4 Ring 3: External Stakeholders (Customers)

This ring represents the wider healthcare ecosystem that interacts with the system via data exchanges.

- **Beneficiaries:** Patients who book appointments and access their medical history.

- **External Partners:** Pharmacies, Insurance Companies, and Laboratories that receive system-generated data.

3.5 Ring 4: Executive Stakeholders (Clients)

The outermost layer consists of decision-makers and authorities with strategic interest in the system.

- **Leadership:** The Healthcare Director and Board of Directors, focused on efficiency and strategic alignment.
- **Regulation:** The Ministry of Health and Government Regulators ensuring legal compliance and public health data reporting.

4. Primary Use Case Workflows

The functionality of the HCMS is defined by core workflows that dictate the interaction between actors and the system.

Use Case Name	Primary Actor	Brief Description
Register New Patient	Receptionist	Captures patient data, validates inputs, and generates a unique Patient ID for clinical use.
Search Patient	Receptionist	Facilitates the retrieval of patient data by name or ID for record updates or appointment management.
Schedule Appointment	Receptionist	Checks doctor availability and allocates time slots to create a verified patient visit.
Add Diagnosis	Doctor	Records medical consultations, treatment notes, and diagnoses directly into the digital record.
Secure Login	All Users	Authenticates credentials to ensure secure access to clinical and operational dashboards.

5. Conclusion

The HCMS provides a comprehensive solution for healthcare management by addressing the needs of patients, providers, and administrators. By centralizing operations and defining rigorous use case workflows, the system architecture ensures that clinical

usability is maintained alongside high standards for data security and regulatory compliance.